

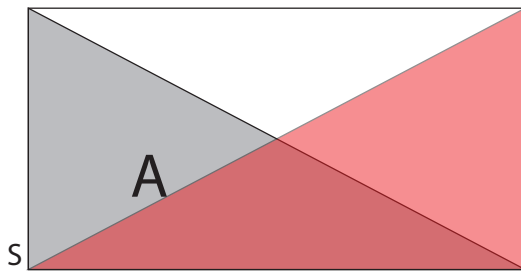
Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE 2500: Engineering Probability, Spring 2026
Exam 1 Solutions

The boxed numbers beside each problem give the mean score for each problem. If your score on a problem is lower than the mean, you need to learn the concept better. The overall mean and median were both 57. There were 6 scores in the 90's.

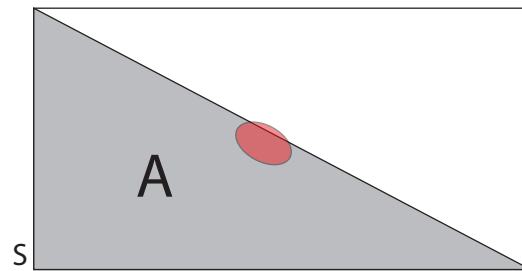
1. (4 points each.) In each picture below, we show the sample space S of an experiment (the outer rectangle) and an event A (the shaded triangle). The distribution of outcomes over the sample space is uniform, so $P(A) = \frac{1}{2}$. In each of the problems below, draw an event B that is **not a subset** of A and satisfies the given requirements. Provide a short explanation for full credit.

12.4

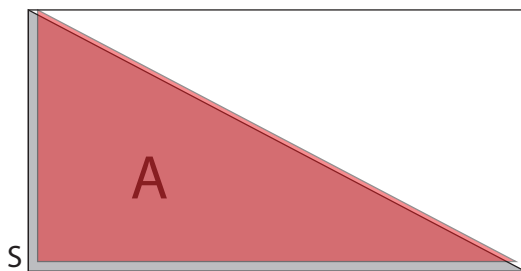
Answers will vary but below are examples that satisfy the stated requirements, where the event B is the red region.



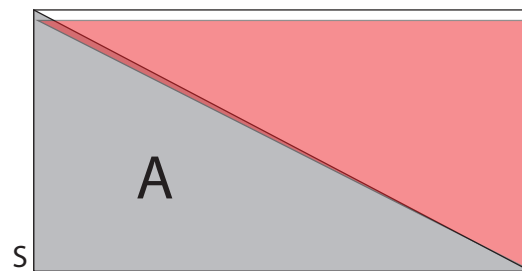
(a) A and B are independent.



(b) $P(B | A)$ is very small but $P(A | B)$ is very large.



(c) $P(B) \approx P(A)$ and $P(B | A)$ is very large.



(d) $P(B) \approx P(A)$ and $P(B | A)$ is very small.

A common error on part (a) was to draw an event B that didn't intersect A at all, and claim the two were independent- but this is not the definition of independence. In this case $P(A \cap B) = 0$ but we would need it to be $P(A)P(B) = \frac{1}{2}P(B)$ for independence. In a correct answer, A should take up exactly half of the event B .

2. (25 points.) Cassian Andor visits a sequence of planets looking for droid friends. Suppose that a droid appears on each planet independently with probability $\frac{1}{4}$. Let X be the random variable corresponding to the planet where Andor encounters the **second droid**.

14.2

- (a) (10 points.) Determine a closed-form formula for the probability mass function of X . Be as precise as possible.

Clearly Andor has to visit at least $k = 2$ planets. If he visits $k > 2$ planets, that means he has to find exactly one droid in the first $k-1$ planets (there are $k-1$ possibilities for where that droid could be, and each has probability $\left(\frac{3}{4}\right)^{(k-2)} \left(\frac{1}{4}\right)$) and then find the second droid on the k^{th} planet with probability $\frac{1}{4}$. So this is kind of like (but not exactly) a geometric random variable:

$$p_X(k) = (k-1) \left(\frac{3}{4}\right)^{(k-2)} \left(\frac{1}{4}\right)^2, \quad k \in \{2, 3, \dots, \infty\}$$

There were very few correct answers here, and incorrect reasoning can lead to a similar-looking answer that you probably didn't get much credit for (so please don't ask). For example, many people jumped to a binomial distribution (i.e., seeing 2 successes in k planets) which gives a similar-looking PMF formula of

$$p_X(k) = \binom{n}{2} \left(\frac{3}{4}\right)^{(k-2)} \left(\frac{1}{4}\right)^2$$

but is definitely not right (e.g., if $n = 7$ the above thinking includes many cases where the second droid is not found on the 7th planet).

Other minor comments are that it's important to clearly indicate the range of k in the PMF formula, and not to assume things we didn't talk about in class (e.g., some people noted correctly that this is called a negative binomial distribution and jumped right to a formula, but we didn't talk about this at all in our lectures/homeworks).

- (b) (5 points.) Compute the decimal value of $P(X = 6)$.

Applying the above formula gives

$$P(X = 6) = p_X(6) = 5 \left(\frac{3}{4}\right)^4 \left(\frac{1}{4}\right)^2 = \frac{405}{4096} = 0.0989$$

You got most of the credit if you proceeded from an incorrect PMF in part (a). You got points deducted if you computed $P(X \leq 6)$ (even if you computed $P(X = 6)$ on the way).

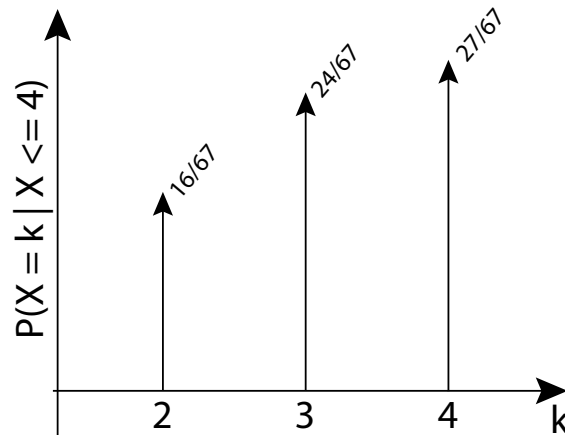
- (c) (5 points.) Compute all values for, and produce an accurate sketch of, $p_X(x \mid X \leq 4)$.

There are only 3 possibilities for $X < 4$, i.e., $X \in \{2, 3, 4\}$. We can compute these as

$$\begin{aligned} P(X = 2) &= \left(\frac{1}{4}\right)^2 &= \frac{1}{16} = \frac{16}{256} \\ P(X = 3) &= 2 \cdot \frac{3}{4} \cdot \left(\frac{1}{4}\right)^2 &= \frac{3}{32} = \frac{24}{256} \\ P(X = 4) &= 3 \left(\frac{3}{4}\right)^2 \left(\frac{1}{4}\right)^2 &= \frac{27}{256} \end{aligned}$$

The total probability of the 3 outcomes is $\frac{67}{256}$ and hence the values of the conditional PMF (sketched below) are:

$$\begin{aligned} P(X = 2 \mid X \leq 4) &= \frac{16}{67} = 0.239 \\ P(X = 3 \mid X \leq 4) &= \frac{24}{67} = 0.358 \\ P(X = 4 \mid X \leq 4) &= \frac{27}{67} = 0.403 \end{aligned}$$



You got full credit for a fully correct process proceeding from a bad PMF in (a). However, the critical element of the problem is normalizing the PMF so it sums to 1. You also lost a point if your PMF has a value at $k = 1$ (Andor can't find 2 droids if he only has visited one planet!). Another common mistake was getting a very high value for the PMF at $k = 4$ by just subtracting the previous two values from 1 without normalizing (which if you think about it, doesn't make much sense).

(d) (5 points.) Compute the decimal value of $E(X \mid X \leq 4)$.

From the above values we can compute that

$$\begin{aligned} E(X) &= 2 \left(\frac{16}{67} \right) + 3 \left(\frac{24}{67} \right) + 4 \left(\frac{27}{67} \right) \\ &= \frac{212}{67} \\ &= 3.164 \end{aligned}$$

You got full credit for a fully correct process proceeding from a bad conditional PMF in (c), even if the resulting value clearly didn't make sense. For example, if your probabilities in (c) don't add up to 1, the expected value you compute is going to be much lower than any of the outcomes! It's good to develop some "does this make sense?" intuition when looking at your answers.

Many answers were incorrect due to not applying the fundamental process of computing expected value (i.e., weighting outcomes by their probabilities and adding up). Even if you mistakenly assumed the random variable was geometric back in (a), you can't apply the $\frac{1}{p}$ expected value formula to the conditional PMF.

3. (25 points.) Suppose we are playing the video game Hades, in which we go through a series of rooms. In each room, a single Greek god appears to give us a gift. Suppose we are required to take gifts from **exactly four** gods. There are 12 gods, and each subset of 4 gods you can get gifts from is equally likely.

12.0

- (a) (5 points.) What is the probability that **both** Artemis and Poseidon are in your chosen set of 4 gods?

First note that there are $\binom{12}{4} = 495$ ways of choosing 4 gods out of the 12. If Artemis and Poseidon are both in the set, that leaves $\binom{10}{2} = 45$ ways of choosing the other 2 gods. So the probability is $\frac{45}{495} = \frac{1}{11}$.

- (b) (5 points.) What is the probability that **exactly one** of Artemis or Poseidon is in your chosen set of 4 gods?

Suppose Artemis is in our set. Then there are $\binom{10}{3} = 120$ ways of choosing the remaining 3 gods (since we can't choose Poseidon). By symmetry there are 120 options containing only Poseidon. So the probability is $\frac{240}{495} = \frac{16}{33}$.

A common, subtle error was to reason that since $P(A) = P(P) = \frac{1}{3}$, then the probability of exactly one of them is $P(A) + P(P) - P(A \cap P) = \frac{19}{33}$. The flaw here is that you would need to subtract off the $P(A \cap P)$ term twice (since both $P(A)$ and $P(P)$ include it, and you want to exclude this possibility).

- (c) (5 points.) Suppose that Artemis is in your set of 4. What is the probability that Poseidon is also in your set?

There are $\binom{11}{3} = 330$ ways of having Artemis in the set, and we computed in part (a) that 45 of these also contain Poseidon. So the probability is $\frac{45}{330} = \frac{3}{11}$. Or there's an even easier way of thinking about it: if Artemis is fixed that means there are 3 remaining slots where Poseidon could be and each has probability $\frac{1}{11}$, giving $\frac{3}{11}$ without any combinatorics.

- (d) (10 points.) Now suppose our set of 4 gods is fixed to be {Artemis, Hera, Poseidon, Zeus}. We allocate 10 offerings to these gods, giving at least 1 offering to each god so nobody feels left out. For example, one allocation could be $(A, H, P, Z) = (3, 2, 3, 2)$. Suppose each of these possible allocations is equally likely. What is the probability that Hera gets more boons than Zeus?

There are three possibilities: $H > Z$, $H = Z$, and $H < Z$. We can see that $P(H > Z)$ has to be the same as $P(Z > H)$ by symmetry, so one way of looking at things is to compute $P(H = Z)$ and then compute $P(H > Z) = \frac{1}{2}(1 - P(H = Z))$.

First compute that the number of possible allocations of 10 offerings to the 4 gods is $\binom{10-1}{4-1} = \binom{9}{3} = 84$. Now consider the ways in which we can have $P(H = Z)$. If Hera and Zeus each have 4 boons, then the other gods have to have 1 boon each and there's only 1 way that can happen. If Hera and Zeus each have 3 boons, then we have 4 boons to distribute to the other gods so they get at least 1 boon each and there are 3 ways that can happen (i.e., $(A = 3, P = 1)$, $(A = 2, P = 2)$, $(A = 1, P = 3)$). We can also apply the hint formula to get $\binom{4-1}{2-1} = 3$. Similarly if Hera and Zeus each have 2 boons, there are 6 boons left for $\binom{6-1}{2-1} = 5$ possibilities and if Hera and Zeus each have 1 boon, there are 8 boons left for $\binom{8-1}{2-1} = 7$ possibilities.

So there are $1 + 3 + 5 + 7 = 16$ favorable outcomes for $H = Z$ over the 84 total outcomes, giving $P(H = Z) = \frac{16}{84}$. Thus $P(H > Z) = \frac{1}{2}(1 - \frac{16}{84}) = \frac{17}{42}$.

4. (18 points.) Prof. Radke tries to sell his photographs at a local art show. The (continuous) dollar amount it costs him to manufacture a small photograph is a uniform random variable on $[10, 30]$ and the (continuous) dollar amount it costs him to manufacture a medium photograph is a uniform random variable on $[30, 60]$. Prof. Radke charges three times his manufacturing cost plus \$10 for each photograph.

Potential customers arrive at his booth according to a Poisson distribution with an average of one customer every 10 minutes. Each customer has probability 0.2 of buying a photograph. If they buy a photograph, they have probability 0.7 of buying a small photograph and 0.3 of buying a medium photograph.

- (a) (10 points.) What is Prof. Radke's expected **profit** over the course of a 6-hour art show?

The expected number of customers N in 6 hours is 36, which is the α for our Poisson random variable. Thus, we expect an average of $36 \cdot 0.2 \cdot 0.7 = 5.04$ small photo sales and an average of $36 \cdot 0.2 \cdot 0.3 = 2.16$ medium photo sales. The expected profit on each small photo sale is $2 \cdot \frac{1}{2}(10 + 30) + 10 = \50 and the expected profit on each medium photo sale is $2 \cdot \frac{1}{2}(30 + 60) + 10 = \100 . Putting it all together, the expected profit is $\$50 \cdot 5.04 + \$100 \cdot 2.16 = \$468$.

- (b) (8 points.) What is the probability that Prof. Radke sells no photographs during a 2-hour event? Be as precise as you can. Hint: remember the Taylor series $\sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x$.

We have to add up all the ways Prof. Radke can sell no photographs: either nobody shows up, 1 person shows up and doesn't buy, 2 people show up and neither buys, and so on. The mean for the Poisson random variable is $\alpha = 12$ expected customers in 2 hours. So we have

$$\begin{aligned}
 P(0 \text{ photos sold}) &= \sum_{k=0}^{\infty} P(0 \text{ photos sold} | k \text{ customers}) P(k \text{ customers}) \\
 &= \sum_{k=0}^{\infty} (0.8)^k \frac{12^k}{k!} e^{-12} \\
 &= e^{-12} \sum_{k=0}^{\infty} \frac{(9.6)^k}{k!} \\
 &= e^{-12} e^{9.6} \\
 &= e^{-2.4} \\
 &= 0.0907
 \end{aligned}$$

where we noticed that the infinite sum for e^x appears in our answer.

5. (16 points.) Jan Dolski collects two types of resources on the surface of a hostile planet: Metals and Organics. Suppose the (integer) number of Metals M_i he collects on day i is a discrete uniform random variable on the range $\{2, 3, 4\}$ and the number of Organics O_i he collects on day i is a binomial random variable with $N = 20$ and $p = \frac{4}{5}$. All of the $\{M_i\}$ and $\{O_i\}$ are independent. Let $M = M_1 + M_2 + M_3 + M_4 + M_5$ and $O = O_1 + O_2 + O_3 + O_4 + O_5$ (that is, the total amount of each resource Jan collects after 5 days).

9.4

- (a) (6 points.) Determine $E(O)$ and $\text{Var}(O)$ using properties of expected value and variance.

$$\begin{aligned} E(O) &= E(O_1 + O_2 + O_3 + O_4 + O_5) \\ &= \sum_{i=1}^5 E(O_i) \\ &= 5 \cdot 20 \cdot \frac{4}{5} \\ &= 80 \end{aligned}$$

$$\begin{aligned} \text{Var}(O) &= \text{Var}(O_1 + O_2 + O_3 + O_4 + O_5) \\ &= \sum_{i=1}^5 \text{Var}(O_i) \\ &= 5 \cdot 20 \cdot \frac{4}{5} \cdot \frac{1}{5} \\ &= 16 \end{aligned}$$

Note that we can simply add the variances since all the random variables are independent.

- (b) (10 points.) Suppose Jan needs 20 Metals to build a battery for his suit. Let $S = M - 20$ be the surplus number of Metals he would have at the end of day 5 (note that S could be negative). Determine $E(S)$ and $\text{Var}(S)$ using properties of expected value and variance.

$$\begin{aligned} E(S) &= E(M - 20) \\ &= \sum_{i=1}^5 E(M_i) - 20 \\ &= 5 \cdot 3 - 20 \\ &= -5 \end{aligned}$$

$$\begin{aligned} \text{Var}(S) &= \text{Var}(M_1 + M_2 + M_3 + M_4 + M_5 - 20) \\ &= \text{Var}(M_1 + M_2 + M_3 + M_4 + M_5) \\ &= \sum_{i=1}^5 \text{Var}(M_i) \\ &= 5 \cdot \frac{3^2 - 1}{12} \\ &= \frac{10}{3} \end{aligned}$$